

A proof of the conjectured recurrence for OEIS A162972

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Abstract

OEIS A162972 carries a conjectured linear recurrence with polynomial coefficients, of order 7. We prove it. The entry's exponential generating function is transcendental, so the usual algebraic arguments do not apply; instead we observe that it lies in a finitely generated module over $\mathbb{Q}(x)$ spanned by monomials $\log(u)^a \exp(g)^k$, and that this module is closed under differentiation, because $(\log u)' = u'/u$ and $(\exp g)' = g' \exp g$ lower the log power and fix the exponential respectively. Re-indexing the recurrence forward turns shifts into derivatives, so the sequence $b(m) = \sum_j q_j(m) a(m+j)$ has exponential generating function $B = \sum_j q_j(\theta) A^{(j)}$, computed exactly inside that module. Here B collapses to the polynomial 0, of degree 0, which proves the recurrence for every $n > 0$.

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1 The sequence and the conjecture

OEIS A162972 is “Number of cycles in all non-derangement permutations of $\{1,2,\dots,n\}$ ”. It has offset 1 and begins

$$a(1), \dots, a(8) = 1, 2, 9, 38, 210, 1339, 9870, 82368, \dots$$

The entry gives the exponential generating function

$$A(x) = \sum_{n \geq 1} a(n) \frac{x^n}{n!} = \frac{(-x - (1 - e^x) \log(1 - x)) e^{-x}}{x - 1}, \quad (1)$$

and it carries the following comment:

Conjecture D-finite with recurrence $a(n) + (-3n+8)a(n-1) + (n-3)(3n-14)a(n-2) + (-n^3+19n^2-98n+156)a(n-3) + (-4n^3+56n^2-258n+395)a(n-4) + (-6n^3+84n^2-395n+624)a(n-5) - (n-5)(4n^2-41n+106)a(n-6) - (n-5)(n-6)^2a(n-7) = 0$. - R. J. Mathar, Jul 26 2022

As of the “Last modified” line on the live entry (2022-07-26, revision 15) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^7 p_i(n) a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = 1, \quad p_1(n) = 8-3n, \quad p_2(n) = (n-3)(3n-14), \quad p_3(n) = -n^3+19n^2-98n+156, \quad p_4(n) = -$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n \geq 0} a(n) x^n / n!$ be the exponential generating function of a sequence, and let $\theta = x \frac{d}{dx}$, so that θ acts on x^m as multiplication by m . For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m) a(m) x^m / m!$.

Lemma 1. *Let $p_0, \dots, p_r$ be polynomials and suppose the conjecture asserts $\sum_{i=0}^r p_i(n)a(n-i) = 0$. Re-index by $n = m + r$ and $j = r - i$, and write $q_j(m) = p_{r-j}(m + r)$, so the claim reads $\sum_{j=0}^r q_j(m)a(m+j) = 0$. Put $b(m) = \sum_j q_j(m)a(m+j)$. Then*

$$B(x) := \sum_{m \geq 0} b(m) \frac{x^m}{m!} = \sum_{j=0}^r (q_j(\theta) A^{(j)})(x),$$

where $A^{(j)}$ denotes the j -th derivative of A .

Proof. For an exponential generating function an upward shift is differentiation: $\sum_{m \geq 0} a(m+j)x^m/m! = A^{(j)}(x)$. Multiplying the coefficient of $x^m/m!$ by $q_j(m)$ is exactly the action of $q_j(\theta)$, and summing over j gives the statement. The re-indexing is what makes this work: a downward shift $a(n-i)$ would correspond to repeated integration, whereas an upward shift is a derivative, and the field used below is closed under differentiation. $\square$

Corollary 2. *With the notation of Lemma 1, the recurrence $\sum_i p_i(n)a(n-i) = 0$ holds for every $n > d$ if and only if B is a polynomial of degree at most d .*

Proof. $b(m)$ is $m!$ times the coefficient of x^m in B , so $b(m) = 0$ for all $m > d$ says exactly that B has no terms above x^d . $\square$

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A , and one only has to recognise it.

Lemma 3. *Let $u, g \in \mathbb{Q}(x)$ and let M be the $\mathbb{Q}(x)$ -module spanned by the monomials $\log(u)^a \exp(g)^k$ for $a \geq 0$ and $k \in \mathbb{Z}$. Then M is closed under d/dx , and hence under $\theta = x d/dx$; explicitly,*

$$\frac{d}{dx} \left(c \log(u)^a \exp(g)^k \right) = \left(c' + c k g' \right) \log(u)^a \exp(g)^k + c a \frac{u'}{u} \log(u)^{a-1} \exp(g)^k, \quad c \in \mathbb{Q}(x).$$

Proof. Differentiate the product. The factor $\exp(g)^k = \exp(kg)$ contributes kg' and does not change the monomial; the factor $\log(u)^a$ contributes $a(u'/u) \log(u)^{a-1}$, lowering the log exponent by one. Both new coefficients lie in $\mathbb{Q}(x)$, so the result is again a finite combination of the same monomials. $\square$

Two features make this usable. The exponential part never mixes monomials at all, and the logarithmic part only ever moves *downwards*, so the module generated by any single element is finite dimensional and every computation terminates. Writing B in this basis, the test of Corollary 2 becomes: B is a polynomial precisely when every coefficient outside the monomial $\log(u)^0 \exp(g)^0$ vanishes and the remaining coefficient has constant denominator. Both are decided by exact cancellation of rational functions, so no series truncation enters anywhere.

3 The computation for A162972

The generating function (1) lies in the module M of Lemma 3. Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K by Lemma 3, the $\sqrt{D}$ component of B cancels identically and the rational part collapses to

$$B(x) = 0,$$

a polynomial of degree 0.

